Purely hydrodynamic origin for swarming of swimming particles

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Three-dimensional simulations with fully resolved hydrodynamics are performed to study the collective motion of model swimmers in bulk and confinement. Calculating the dynamic structure factor, we clarified that the swarming in bulk systems can be understood as a pseudoacoustic mode. Under confinement between flat parallel walls, this pseudoacoustic mode leads to a traveling wavelike motion. This swarming behavior is due purely to the hydrodynamic interactions between the swimmers and depends strongly on the type and strength of swimming (i.e., pusher or puller).

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I. INTRODUCTION

Active matter systems such as groups of microorganisms, birds, or fish often form swarms or flocks and typically show complex collective behavior [1–3]. Of special interest for scientific study are active microscopic systems, e.g., microorganisms or Janus particles, because they are particularly well suited for experiments. Understanding the collective dynamic behavior of these systems can be useful for both science and industrial applications, for example, to explain the formation of biofilms [4] or to design targeted drug delivery systems [5]. Such systems typically exist under some type of confinement, which has been shown to exert a strong influence on the collective behavior [6–10]. So far, most analyses of such systems have either neglected hydrodynamic interactions or only considered them in a far-field approximation [6,9,11,12], even though it is known that the hydrodynamic interactions can dramatically affect the physical properties of the systems [13]. Theoretically, it is impossible to take into account the full hydrodynamic interactions in many-body systems, and numerically it is very computationally expensive. While several studies have succeeded in including hydrodynamic interactions [14–16], only few have focused on the collective dynamic behavior of three-dimensional (3D) active systems under confinement.

We have conducted direct numerical calculations for 3D systems of spherical swimmers in a host viscous fluid with fully resolved hydrodynamics and provide three novel findings. First, we elucidated that the swarming behavior in bulk puller dispersion, which Alarcon and Pagonabarraga reported [15], can be understood as a pseudosound mode, as evidenced by the dynamic structure factor. Second, under the confinement of two flat parallel walls, we find collective motion of swimmers with traveling wave characteristics, which propagates back and forth between the walls. Although similar traveling wavelike motion has already been observed in systems with explicit alignment interactions between the particles [17,18], here we show that hydrodynamic interactions alone are enough to exhibit this behavior. Third, we show evidence that this traveling wavelike behavior can be regarded as a manifestation of the pseudo-sound mode of the system. Crucial to observe this wavelike behavior are the strength and type (pushers vs pullers) of swimming, not the degree of global alignment.

II. SIMULATION METHODS

A. The squirmer model

As the numerical model of microswimmers, the squirmer model was employed. In this model, a self-propelled particle is modeled as a spherical object with a modified stick boundary condition at its surface [19,20]:

$$\boldsymbol{u}^s(\hat{\boldsymbol{r}}) = \sum_{n=1}^{\infty} \frac{2}{n(n+1)} B_n P_n'(\cos\theta)\sin\theta\hat{\boldsymbol{\theta}}, \tag{1}$$

where $\hat{\boldsymbol{r}}$ is a unit vector directed from the center of a squirmer to a point on its surface, $\theta = \cos^{-1}(\hat{\boldsymbol{r}}\cdot\hat{\boldsymbol{e}})$, the polar angle between $\hat{\boldsymbol{r}}$ and the swimming direction $\hat{\boldsymbol{e}}$, $\hat{\boldsymbol{\theta}}$ the unit polar angle vector. P_n' is the derivative of the Legendre polynomial of nth order, and B_n is the magnitude of each mode. Here the radial deformation has been ignored, so this surface velocity has only the tangential component and is responsible for the self-propulsion of the swimmers.

In this work, only the first two modes in Eq. (1) were retained:

$$\boldsymbol{u}^s(\theta) = B_1\left(\sin\theta + \frac{\alpha}{2}\sin 2\theta\right)\hat{\boldsymbol{\theta}}. \tag{2}$$

The coefficient of the first mode, B_1, determines the swimming velocity of an isolated squirmer ($U_0 = 2/3B_1$). The coefficient of the second mode, B_2, determines the stress exerted by the particles on the fluid [21]. Using only the first two modes, this type of swimming can be modified by changing the sign of the $\alpha = B_2/B_1$ coefficient in Eq. (2). Negative values of α describe pushers, which generate an extensile flow field in the propelling direction, and positive values describe pullers, which generate a contractile flow (Fig. 1). Because these two swimming mechanisms result in distinct flow fields, different swimmers can exhibit vastly different collective motion, as we will show in this paper.

B. Smoothed profile method

In this study, the squirmer model is incorporated into the smoothed profile method (SPM) [22–24], which is an efficient calculation scheme for solid-fluid two-phase dynamics problems with full hydrodynamics. In this method, the sharp boundary between the solid and fluid domains is not considered explicitly. Instead, a diffuse interface of finite width ζ is

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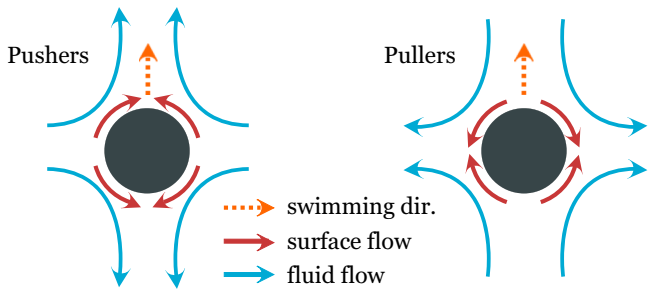

FIG. 1. Schematic representation of pushers and pullers within the squirmer model. The swimming direction and the fluid and surface flows generated by squirmers are shown.

introduced, and the solid domain is represented via an order parameter ϕ, which takes a value of 1 in the solid domain and 0 in the fluid domain. Using this continuous order parameter, all the physical quantities can be expressed as field variables on Cartesian grids, and the calculation cost can be reduced drastically. As the governing equations for the total fluid, we employ a modified incompressible Navier-Stokes equation:

$$\begin{aligned} \rho_f(\partial_t + \boldsymbol{u}\cdot\nabla)\boldsymbol{u} &= \nabla\cdot\boldsymbol{\sigma} + \rho_f(\phi\boldsymbol{f}_p + \boldsymbol{f}_{\mathrm{sq}}),\\ \nabla\cdot\boldsymbol{u} &= 0,\\ \boldsymbol{u} &= (1-\phi)\boldsymbol{u}_f + \phi\boldsymbol{u}_p,\\ \phi\boldsymbol{u}_p &= \sum_i \phi_i[\boldsymbol{V}_i + \boldsymbol{\Omega}_i\times\boldsymbol{R}_i], \end{aligned} \tag{3}$$

where $\boldsymbol{u}$ is the total velocity field and $(1-\phi)\boldsymbol{u}_f$ and $\phi\boldsymbol{u}_p$ the velocity field of fluid and particles, respectively. The position, velocity, and angular velocity of particle i are expressed as $\boldsymbol{R}_i$, $\boldsymbol{V}_i$, and $\boldsymbol{\Omega}_i$. The stress tensor and the mass density of the host fluid are $\boldsymbol{\sigma}$ and ρ_f. The body force generated from the rigidity condition of particles is expressed as $\phi\boldsymbol{f}_p$, and $\boldsymbol{f}_{\mathrm{sq}}$ stands for the force distribution resulting from the squirming motion (the effects due to the slip-velocity $\boldsymbol{u}^s$ are contained in $\boldsymbol{f}_{\mathrm{sq}}$) [25]. The time evolution of the particles follows from the Newton-Euler equations:

$$\begin{aligned} \dot{\boldsymbol{R}}_i &= \boldsymbol{V}_i, \quad \dot{\boldsymbol{Q}}_i = \mathrm{skew}(\boldsymbol{\Omega}_i)\cdot\boldsymbol{Q}_i,\\ M_p\dot{\boldsymbol{V}}_i &= \boldsymbol{F}_i^H + \boldsymbol{F}_i^C, \quad \boldsymbol{I}_p\cdot\dot{\boldsymbol{\Omega}}_i = \boldsymbol{N}_i^H, \end{aligned} \tag{4}$$

where $\boldsymbol{Q}_i$ denotes the rotation matrix, M_p the mass, and $\mathrm{skew}(\boldsymbol{\Omega}_i)$ the skew symmetric matrix of $\boldsymbol{\Omega}_i$, and $\boldsymbol{F}_i^H$ and $\boldsymbol{N}_i^H$ are the hydrodynamic force and torque acting on particle i. The force due to direct interactions between particles, $\boldsymbol{F}_i^C$, is taken to be a truncated Lennard-Jones potential to ensure that the excluded volume constraint is satisfied. See Refs. [22–25] for more detailed information on this method.

III. RESULTS

A. Bulk behaviors

We conducted simulations of swimmer dispersion systems in a three-dimensional domain. Throughout this paper, the diameter a and the boundary thickness ζ of the squirmers are 4Δ and 2Δ, respectively, where Δ is the grid spacing and the unit length. The B_1 parameter in Eq. (2) is set as 0.375. The shear viscosity η and the fluid mass density ρ_f are set to 1, and the unit time is then expressed as $\rho\Delta^2/\mu$. The particle Reynolds number of an isolated squirmer $\mathrm{Re}^0 = \rho_f U_0 a/\eta$ is set equal to 1. The Reynolds number in the dispersion will be different from Re^0 due to a decrease in the propelling velocity with increasing volume fraction. We have ignored any effects due to thermal fluctuations. No external force is exerted, and the systems are buoyancy free.

We first calculated the dynamics of swimmers in bulk cubic systems. The linear dimension of the computational domain is $L = 64\Delta$ and the number of swimmers is 1043 (volume fraction of 13%). Periodic boundary conditions are set in every direction. We conducted simulations for various α values in order to investigate the effect of α on the dynamics. Alarcon and Pagonabarraga have considered a similar setup and showed that puller suspensions with $\alpha = 0.5$ can exhibit swarming behavior [15]. The swarming is purely hydrodynamic in origin, as no direct aligning interactions are included. In Ref. [15], the swarming behavior was characterized in terms of the local structure (generalized radial distribution functions and the local density fluctuations), but no attempt was made to quantitatively analyze the collective dynamic behavior of the system. To accomplish this, we have calculated the dynamic structure factor $S(\boldsymbol{k},\omega)$, which is just the Fourier transform of the density-density correlation function in both space and time [26]:

$$\rho_{\boldsymbol{k}}(t) = \sum_{j=1}^{N} \exp[-i\boldsymbol{k}\cdot\boldsymbol{R}_j(t)] \tag{5}$$

$$S(\boldsymbol{k},\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{N}\langle\rho_{\boldsymbol{k}}(t)\rho_{-\boldsymbol{k}}(0)\rangle\exp(i\omega t)dt, \tag{6}$$

where $\boldsymbol{k}$ is the wave vector and ω the angular frequency.

The spherically averaged (over shells of wave vectors) structure factors of the systems for $\alpha = \pm 0.5$ are shown in Fig. 2. Shifted peaks were observed for both systems, pullers ($\alpha = 0.5$) and pushers ($\alpha = -0.5$). These peak positions ω_{shift} follow a Brillouin-like dispersion relation:

$$\omega_{\mathrm{shift}} = c|\boldsymbol{k}|, \tag{7}$$

where c is a coefficient which has the dimension of velocity (in the case of normal Brillouin peaks, c represents the sound velocity). We can therefore say that these systems possess pseudoacoustic mode dynamics or a soundlike propagation mode for the density. No Rayleigh-like peaks were observed for small wave vectors, probably because the Brillouin-like peaks are much stronger (for wave vectors of large magnitude, Rayleigh-like peaks are recovered). The sound velocity of this pseudoacoustic mode can be calculated from the dispersion relation, shown at the bottom of Fig. 2 and Eq. (7) [note that the values in Fig. 2(c) are all nondimensionalized]. We do not see any difference between the sound velocities of pushers and pullers, although the peak intensities show very big differences.

B. In confinement

As we show below, the pseudosound behavior presented above can have dramatic consequences for systems under confinement, which is relevant to most experimental realizations of active matter systems. We have calculated the dynamics

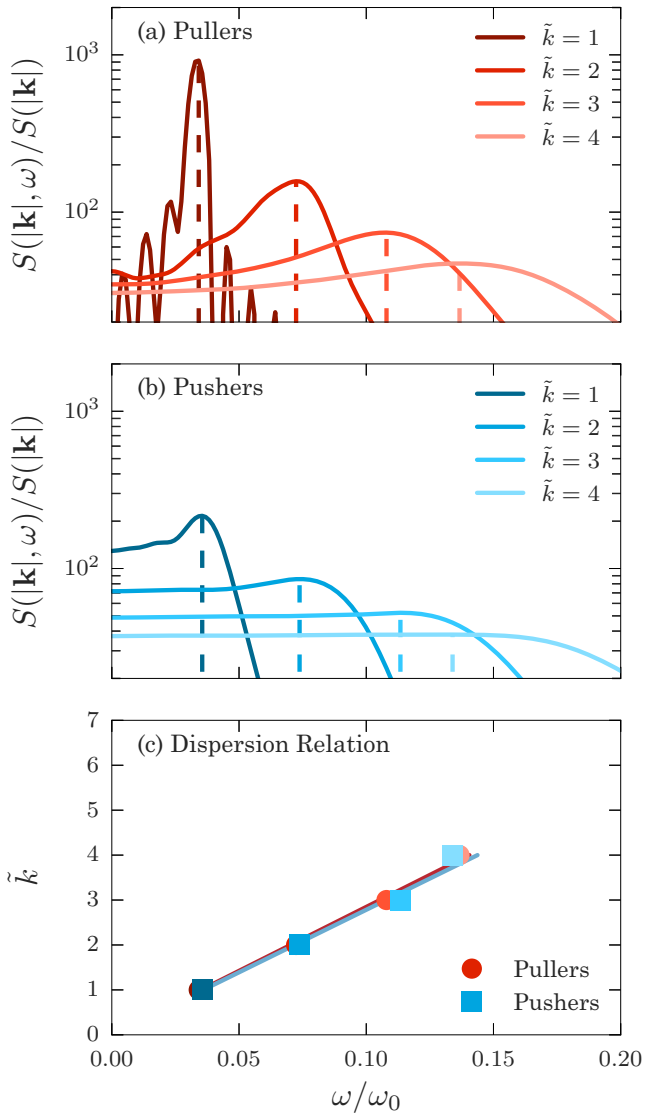

FIG. 2. The calculation results for the spherically averaged dynamic structure factor (DSF) $S(\boldsymbol{k},\omega)$, where $\boldsymbol{k} = \frac{2\pi}{L}\tilde{\boldsymbol{k}}$ is the wave vector and ω the angular frequency (normalized by $\omega_0 = 2\pi/t_0$, where $t_0 = a/U_0$). The top two figures show the simulation results for the DSF of (a) pullers at $\alpha = 0.5$ and (b) pushers at $\alpha = -0.5$. The bottom figure (c) shows the relation between wave vectors and the positions of the Brillouin peaks. Arrows in (a) and (b) show the positions of peaks.

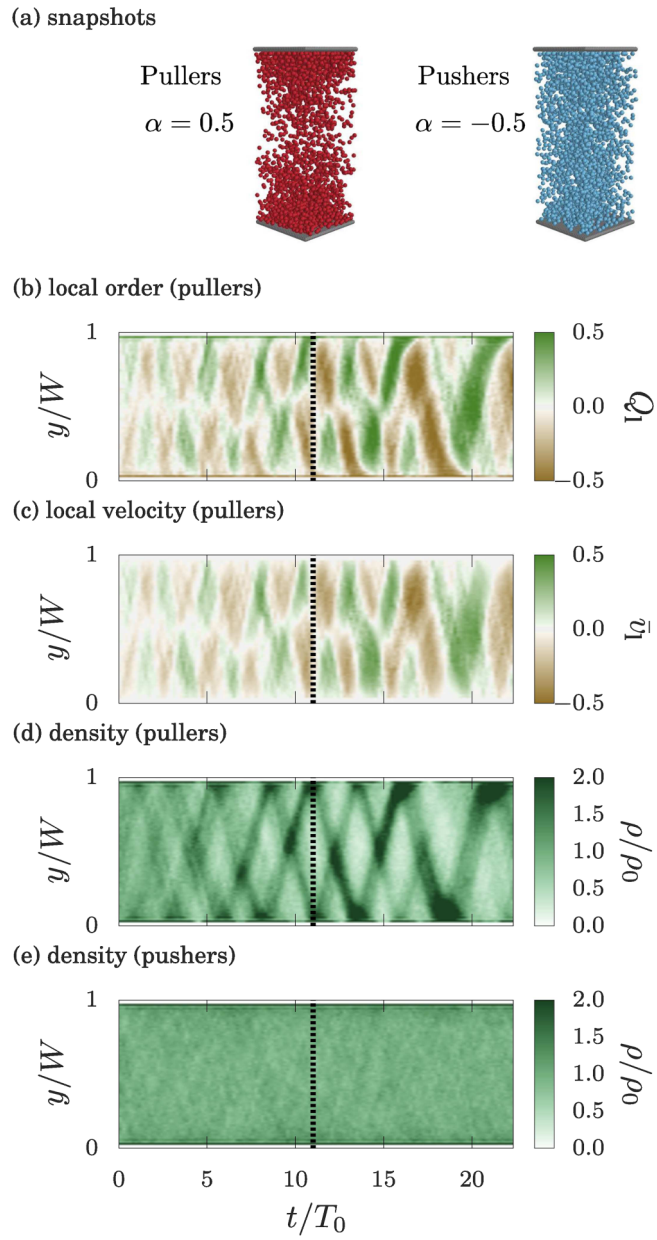

FIG. 3. Simulation results for pullers and pushers ($\alpha = \pm 0.5$) at volume fraction of 13% and wall separation $W/a = 40$. (a) Simulation snapshots, (b) local order of pullers defined as $Q_1 = \langle \hat{e}_y \rangle_{\text{plane}}$, (c) local velocity of pullers in y direction (perpendicular to the walls) defined as $\bar{v}_1 = \langle V_y/U_0 \rangle_{\text{plane}}$ (d) and (e) plane averaged density for pushers and pullers at each height y/W. The density is normalized by the global average value, ρ_0, and time and velocity are normalized by $T_0 = W/U_0$ and U_0, where W represents the separation of parallel flat walls. The snapshots are of around $t = 11T_0$, which is marked by dashed line.

of $N = 2607$ swimmers in a $64\Delta \times 160\Delta \times 64\Delta$ rectangular system (volume fraction of 13%, the same value as the previous simulations in the bulk system). The computational domain is periodic in the x and z directions but confined between two flat parallel walls along the y direction; at $y = 1\Delta$ and 159Δ. These walls are made of 2048 spherical particles which are pinned and unable to move or rotate from the initial positions and orientations. The wall particles are regularly spaced and are defined to have the same diameter as the squirmers. The initial configuration of swimmers was randomly determined with respect to both position and orientation. We again conducted simulations for various values of α. We have calculated the time evolution of the local density ρ, polar order Q_1, and velocity v_1 as a function of distance from the walls. Here $Q_1 = \langle \hat{e}_y \rangle_{\text{plane}}$ and $\bar{v}_1 = \langle V_y/U_0 \rangle_{\text{plane}}$, where $\hat{e}_y$ and V_y are the components of $\hat{\boldsymbol{e}}$ and $\boldsymbol{V}$ perpendicular to the walls, and $\langle \cdot \rangle_{\text{plane}}$ denotes an ensemble average over all particles in parallel slabs of width 2Δ. Figure 3 shows the results of the systems with $\alpha = \pm 0.5$ (see movies S1 and S2 in the Supplemental Material [27]). For pullers ($\alpha = 0.5$), the

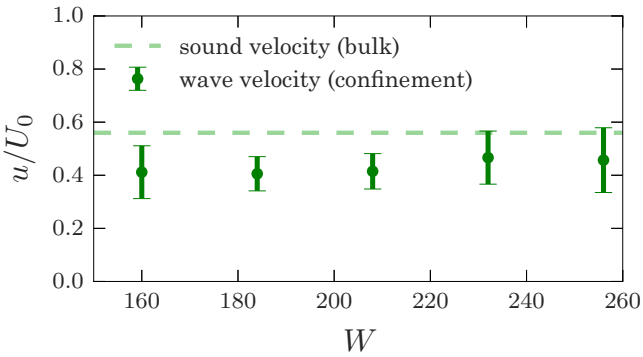

FIG. 4. The traveling wave velocity of confined systems as a function of wall separation W. The sound velocity of the bulk system is also shown. Here, only puller results are shown, as pushers do not show wavelike behavior. All the values are normalized by U_0.

time evolution of the density, polar order, and velocity are shown, while for pushers ($\alpha = -0.5$), only density is shown. Only the pullers show large density fluctuations; pushers show no such behavior. First, the pullers build up two distinct traveling waves which bounce back and forth between the walls. Gradually, as one wave begins to dominate, they merge into a single propagating wave. We also considered different wall separations and found that such propagating wavelike motion can only be observed if the wall separation is larger than the characteristic size of this propagating flock. In our simulations, the threshold is $W/a \simeq 40$, where W is the wall separation. In Fig. 4, the traveling wave velocity is plotted for several different wall separations. They are quantitatively the same and there is no clear separation dependence (see Ref. [28] for the wave velocity measurement scheme). The value of the sound velocity for the pullers calculated from the dispersion relation is also shown. This pseudosound velocity is quantitatively the same as the wave propagating velocity in confinement, though slightly higher. Thus, we believe that this traveling wavelike behavior under confinement is due mainly to the intrinsic pseudoacoustic properties of the swimmers. In this case, the role of the walls is to break the symmetry and provide a finite domain over which the density waves can develop.

We have performed simulations for volume fractions up to 30%, with smaller and larger wall surface areas and different α values ($-5 \leqslant \alpha \leqslant 5$). The traveling wavelike motion is always observed for pullers in the range $0 < \alpha < 1$.

Although it is known that pullers can show swarming behavior in bulk [15], the traveling wavelike motion under confinement we present here was unexpected and surprising for the following two reasons. First, the walls can strongly affect the hydrodynamic interactions between particles [6–10]. Actually, for small separations, we see suppression of this traveling wavelike motion, while, for sufficiently large separations, walls allow us to see constructive interference. Second, even if the swarming can be observed, it is possible that the particles will prefer to swim parallel to the walls. Indeed, this is exactly what happens in the case of neutral swimmers ($\alpha = 0$) (S3 in Ref. [27]).

IV. DISCUSSION

Now we discuss the origin of the pseudoacoustic mode. From Fig. 3, we can see that the particles show local polar order and move as a group collectively, even though individual particles seem to move independently from their neighbors (for individual particles motion, see movie S4 in Ref. [27]). In puller systems, this cooperative swimming as a group can be understood as the origin of the pseudoacoustic mode or density propagation. Now let us consider a simplified setup in which swimmers are placed along a straight line with a nonregular distribution, and oriented parallel to this line, in such a way they will start swimming in the same direction (S5 and S6 in Ref. [27]; simulations are conducted in the full-3D domain). In this situation, due to the contractile flow along the swimming direction, pullers attract each other and tend to enhance the longitudinal mode of density fluctuations, resulting in the formation of flocks. On the other hand, pushers show the opposite tendency, which suppresses the longitudinal fluctuations. In the systems of interest in this paper, the situation is much more complicated. However, we can still say that the appearance of the large density fluctuations we present depends on the local interaction rather than on the global order. Actually, we never observed large fluctuations for pushers, even though the global polar order ($Q = \langle \frac{1}{N} | \sum_i \hat{\boldsymbol{e}}_i | \rangle$) [14,15] could be similar to pullers with $\alpha = 0.5$. On the other hand, pullers with various values of α ($0.3 \leqslant \alpha \leqslant 0.9$) do show traveling wavelike motion in confinement.

Let us consider another simplified setup, in which the swimmers are placed along a straight line with the same nonregular distribution but oriented perpendicular to the line (see S7a and S8a in Ref. [27]). In the case of pullers, the differences in the interparticle separations, which can be regarded as transverse density fluctuations, are actually suppressed by the hydrodynamic interactions. We can understand this as follows: pullers aligned parallel to each other (side by side) tend to repel themselves due to the contractile flow field they generate. As for pushers, we observe a cyclic bouncing back motion between neighboring particles, in which the separation between the two most distant particles is increased. Thus, for pushers, we observed a kind of transverse instability, although it is much weaker and more short lived compared to the longitudinal one seen for pullers. Both pushers and pullers show segregation of the particles into two distinct line clusters, one behind the other. We consider that this is due to the small interparticle separation. Actually, if we take as a starting configuration for a new simulation one of these line clusters (from the steady state of the puller system), we do not see such separation (see S7b and S8b in Ref. [27]), but we still see a great difference in the density fluctuations between pullers and pushers.

In order to rule out the possibility that this traveling wavelike motion is due to inertial effects, we also conducted simulations with a smaller Reynolds number ($\mathrm{Re}_0 = 0.1$). The system showed the same collective behavior. We can therefore conclude that this behavior is purely hydrodynamic in origin.

As mentioned above, the neutral system ($\alpha = 0$) shows a unique behavior due to absence of the second mode in the right-hand side of Eq. (2). In bulk, a flocking state will soon develop, in which all particles swim in the same direction

with the same velocity. As there are only relatively small changes in their relative positions, the intermediate scattering function will exhibit a very long relaxation time. Therefore, it is practically impossible to calculate the corresponding dynamic structure factors, which are shown in Fig. 2 for pullers and pushers. Instead, we use the average swimmer velocity in the flocking state to define the characteristic velocity of the neutral swimmer dispersions. The average velocities in bulk (volume fraction of 13%) and confinement ($W = 160\Delta$, volume fraction of 13%) are $0.76U_0$ and $0.95U_0$, respectively; the velocity in confinement is considerably larger than that in bulk. Regarding the increased velocity for the neutral swimmers, as shown by Li *et al.* [16], the swimming velocity can be both increased or decreased due to the presence of a wall (depending on Re and the swimming parameter α). In summary, the dynamics under confinement are qualitatively the same as those of bulk systems, even though the flocking velocity can differ.

We mention previous works using theoretical continuum models which have shown similar results, although it is difficult to make a one-to-one comparison with our work. Simha and Ramaswamy predicted, using a continuum model, that active matter suspensions with polar order can show sound like propagation modes [11]. Ezhilan *et al.* studied a similar continuum model [12]. They reported that pushers show a stronger instability in their concentration field, contrary to our results that pullers have much higher Brillouin-like peaks and are more likely to show density fluctuations. However, these continuum models only include far-field hydrodynamics, whereas we considered the full hydordynamic interactions and the finite size of the particles. As reported in preceding works, the near-field hydrodynamic interactions strongly affect the collective dynamics [29,30]. In particular, most of the theoretical works assume an elongated shape for the particles, which leads to local alignment via steric interactions, while we consider spherical particles.

The recent work by Zöttl and Stark [31] is related to ours, but deals with quasi-2D systems under strong confinement (small separation of walls) and high volume fraction, which are quite different conditions from those of the present work. The values of the swimming strength also crucially differ from ours, focusing mainly on strong pushers and pullers $|\alpha| > 1$ Although they have also considered neutral particles $\alpha = 0$, they have not considered any pusher-puller system with $|\alpha| < 1$, which is where we observe the traveling wavelike motion (as we have mentioned above, systems with $\alpha = 0$ behave in a completely different manner from systems with $\alpha \neq 0$). Not surprisingly, we are able to observe quite different dynamic behaviors. The works by Lefauve and Saintillan [32] and Tsang and Kanso [33] also refer to traveling wavelike behavior in systems of microswimmers. They reported density shock waves at the front and back of the flocks, which we could not observe in our simulations. This is likely due to the addition of the background flow and strong Hele-Shaw-type confinement used in these works. Actually, Desreumaux *et al.* reported that even passive colloids can show shock waves when they are driven by a background flow [34].

V. CONCLUSION

In conclusion, we confirmed, using direct numerical simulations with full hydrodynamics, that swimmers can show pseudoacoustic properties in bulk. Also, swimmer dispersions confined between flat parallel walls can exhibit a traveling wavelike motion. Furthermore, the propagating velocity of this traveling wave does not depend on the wall separation, and this value is in quantitative agreement with the pseudosound velocity in bulk. Judging from these results, the presence of the walls does not strongly affect the dynamics. Rather, they just play a passive role, breaking the symmetry of the system and providing a finite domain over which the density wave can develop, thus allowing the pseudoacoustic property to manifest itself.

This kind of phenomenon cannot be expected in conventional passive colloidal dispersions, since such density fluctuations would be suppressed due to viscous damping [35]. In future works, we will consider complex wall geometries, such as gear-shaped objects [6,9] and circular systems [8]. Based on our current results, we expect that spherically shaped particles, without any apparent alignment interaction, can generate nontrivial collective motion due solely to hydrodynamic interactions.

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[27] See Supplemental Material at http://link.aps.org/supplemental/10.1103/PhysRevE.93.043114 for the simulation movies.
[28] To calculate the wave velocity, we proceeded as follows. First, we divide the system into slabs of thickness $0.5a$ parallel to the walls and calculate the local order parameter, Q_l, and perpendicular velocity, $\bar{v}_l$, in each slab. In Fig. 2, we show Q_l, $\bar{v}_l$, and ρ as functions of height and time. We clearly see evidence of a propagating high density flock, which appears as a dark diagonal band in the density plot. We identify the moving flock by looking for regions with $Q_l > Q_t$ and $\bar{v}_l > v_t$, where Q_t and v_t are the threshold for each variable. The threshold for the local order parameter, Q_t, is set at 0.5 and we consider that the slab is occupied by the flock if Q_l is higher than this value. In addition, we use a threshold v_t for the velocity to remove the data near the wall boundaries where the average local velocity should vanish. We set v_t at $0.04U_0$, which is less than 10% of the wave velocity (or the sound velocity).
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